

Results: 12-month aspiration rates were similar for the training (8/238 = 3.4%) and test cohort (2/88 = 2.3%). Age at RT, baseline PAS, obesity (BMI > 31), pharyngeal constrictor V4Gy, and 12-month EORTC QLQ HN35 swallowing score were positively associated with the likelihood of aspiration based on 95% credible intervals. Model training AUC was 0.97 (95% CI = 0.97-1.0), and test AUC was 0.90 (0.70-1.0). Using the training-optimized predicted probability cut point to predict aspiration on the test cohort, prediction accuracy was 0.97 (0.90-0.99), specificity was 0.98 (0.92-1.0), sensitivity 0.5 (0.01-0.99), PPV 0.33 (0.01-0.91), and NPV 0.99 (0.92-1.0).

Conclusion: OPSCC 12-month post-RT aspiration can be reliably predicted using patient demographics, treatment characteristics, and PROs. The model's test NPV of 0.99, provides a novel tool to tailor the intensity of patient intervention during the surveillance phase. Using this validated model could have reduced the number of 12-month MBSS ordered from 88 to 3 (97% reduction) in the test cohort.

Abstract 3613 — Table 1: Test and training confusion matrices

Test Predicted PAS⇒	Test Reference ⇓PAS⇓		Training Predicted PAS⇒	Training Reference ⇓PAS⇓	
	>=6 <6			>=6 <6	
	>=6	<6		>=6	<6
>=6	1	2	>=6	7	10
<6	1	84	<6	1	220

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Adjuvant Ruthenium Plaque Brachytherapy (BRT) after Partial Lamellar Vitreo-Uvectomy (PLSU) for Medium-Large Uveal Melanoma (UM)

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Purpose/Objective(s): To assess rates local recurrence (LRR), useful vision, complications, distant metastases (DM) and overall survival (OS) in patients treated with PLSU and adjuvant BRT.

Materials/Methods: The medical records of twenty-one patients with UM, not candidates for BRT alone, treated with PLSU and BRT from January 2017 till February 2022, were reviewed. Patients received median radiation dose of 85 Gy (range = 80-100), calculated at 3 mm depth, assuming a 1 mm scleral thickness. Useful vision was defined as VA ≥ 0.2 on the decimal scale.

Results: Median follow-up was 49 months (range = 21-84). Median tumor thickness was 7.3 mm (range = 4.6-12.2) and median tumor largest basal diameter (LBD) was 12 mm (range = 8-18). Eight patients (28.6%) had local recurrence. Seven patients (33.3%) had deterioration in their visual acuity during follow-up and one patient reported improvement in vision. Four patients (19%) underwent enucleation with globe preservation rate of 81%. One patient (4.7%) developed distant metastases during follow-up and one patient died due to unrelated cause. Eight patients (38.1%) developed vitreous hemorrhage, eight

patients (38.1%) had subretinal fluid accumulation and six patients (28.6%) developed cataract.

Conclusion: Adjuvant Ruthenium brachytherapy after PLSU is a good alternative to enucleation in patients who are not candidates for brachytherapy alone with excellent globe preservation rate. However, visual acuity deteriorated in substantial percentage of patients.

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Definitive and Salvage Stereotactic Body Radiation Therapy (SBRT) for Nonmelanoma Skin Cancer of the Head and Neck

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Purpose/Objective(s): Patients with head and neck locoregional only nonmelanoma skin cancer (NMSC) who are not candidates for surgery or immunotherapy, or progress after prior treatment, have limited options. SBRT offers a noninvasive, short, definitive treatment with the potential for durable control.

Materials/Methods: An IRB approved registry of head and neck cancer patients was queried for NMSC treated with 5 fraction SBRT (minimum 35 Gy) from 2014 to present. Acute toxicity (<90 days from SBRT) graded per CTCAE v4.0 was evaluated. Crude rates of locoregional control (LRC) and distant metastasis (DM) were calculated.

Results: Head and neck SBRT was used in 13 patients with NMSC. The median follow up was 9.8 months (range 3.8-57.6) and median age was 73.4 years (range 60.7-91.5). Histologies included squamous cell carcinoma (77%), basal cell carcinoma (15%), porocarcinoma (8%). 15% were immunosuppressed, 31% had prior parotidectomy, and 77% had prior overlapping radiation. Median time from prior radiation to SBRT was 12 months (range 4.1-17.4). Treatment site was neck/parotid in 77%, base of skull 23%. Concurrent and adjuvant cetuximab and immunotherapy was given in 23% and 15%, respectively. SBRT dose was 45 Gy (7.5%), 40 Gy (69%), 37.5 Gy (7.5%), 35 Gy (15%). Secondary elective volume of 25-35 Gy was used in 38.5%. Median high dose and elective PTV volumes were 79 cc (range 7-165) and 276 cc (range 158-631), respectively. Bolus was used in 38.5%. LRC was 100% for patients without prior RT (N=3) and 60% for re-irradiation (N=6/10), with all failures within the high dose region. DM developed in 38.5% (N=5), 2 with synchronous locoregional failure. Acute grade 3+ toxicity was seen in 15% of patients (N=2), which included grade 3 dermatitis (N=1) and grade 3 meningitis (N=1; exposed calvarium present at baseline).

Conclusion: SBRT in the primary or re-irradiation setting offers a short and effective treatment option for patients with NMSC that are not candidates for, or refuse, standard therapy. Longer follow up is needed to assess the incidence of late effects, however this is a promising treatment approach for select patients.

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